TAO Yuhao

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EDUCATION

ETH Zürich, Switzerland Sep. 2023-present

M.Sc. in Robotics, Systems and Control (with distinction) **Grade**: 5.84/6.0

- **Core Courses**: Microrobotics, Nanorobotics, Introduction to Machine Learning, Image Analysis and Computer Vision, Dynamic Programming and Optimal Control, Linear System Theory, Recursive Estimation, Biomedical Imaging.

RWTH Aachen University, Germany Oct. 2018-Jul. 2023

B.Sc. in Mechanical Engineering (Graduation with distinction) **Grade**: 1.1/1.0 (Ranking: 2/1885)

- **Awards**: Dean's List 2019/2020, 2020/2021 (Top 5%); Germany Scholarship 2020/2021, 2022/2023
- **Core Courses**: Solid Mechanics (Statics, Elasticity, Dynamics), Material Science, Machine Design, Simulation Methods in Engineering, Automatic Control, Numerical Mathematics, Fluid Mechanics, Thermodynamics, etc.
- **GRE on-site**: 336 (Verbal 166, Quantitative 170) + 3.5 (Writing); **TOEFL on-site**: 112 (Speaking 28)

RESEARCH EXPERIENCE

@ **Multi-Scale Robotics Lab, ETH Zürich** | Supervisor: **Prof. Bradley Nelson** (bnelson@ethz.ch)

Deployable Origami-Antenna Capsule Robot (paper in preparation) Sep. 2024-Sep. 2025

Designed and fabricated a deployable origami capsule robot for small-intestine motility monitoring. Developed the pop-up laminate structure, designed NdFeB magnet spring for magnetic navigation, and developed an Abaqus DLOAD co-simulation platform for magnet interaction. Developed a deformation-coupled 3D meander-line antenna and validated its S11–geometry mapping via HFSS and benchtop measurements. Co-developed and tested a custom flexible PCB using an LMX2594 PLL for GHz frequency sweeps and implemented on-board control firmware. Conducted robot deployment tests, magnetic steering, and endoscopic evaluations in intestine models, and participated in in-vivo porcine trials.

Bioadhesive Patch Robot with Integrated Sensing Dec. 2024-Sep. 2025

Developed a multilayer adhesive-patch robot for reflux monitoring. Fabricated a hydrogel with thickness-dependent lap-shear strength and optimized adhesion on porcine esophagus. Prepared a magnetically loaded PDMS layer enabling remote manipulation. Co-designed a flexible PCB with calibrated pH/T sensors, low-power readout circuitry, and BLE telemetry. Implemented a real-time GUI for data logging and validated end-to-end accuracy via voltage-transmission correction. Demonstrated magnetic navigation, robust mucosal adhesion, and synchronized pH/T sensing.

@ **Institute of Neuroscience, Chinese Academy of Sciences** | Supervisor: **Prof. Zhengtuo Zhao** (zhaozt@ion.ac.cn)

Vision-Guided Robotic Brain Implantation Platform (paper in preparation) Dec. 2025-Mar. 2026

Developed a calibration-free, vision-guided robotic system for marmoset brain reconstruction and electrode implantation, integrating automated multi-image stitching, pixel-to-robot-space mapping, and needle localization to enable ~10 µm precision surgical targeting. Validated the system in marmoset animal trials.

@ **AIA Institute, RWTH Aachen** | Supervisor: **Prof. Wolfgang Schröder** (dekan@fb4.rwth-aachen.de)

Modeling Kolmogorov-Scale Deformable Bubbles in Turbulence Jan. 2023-Apr. 2023

Extended a C++ LPT solver for deformable particles, validated bubble flows, and analyzed turbulence parameters

@ **Institute of Mechanics, Chinese Academy of Sciences** | Supervisor: **Prof. Xiaolei Wu** (xlwu@imech.ac.cn)

Atomistic Deformation Studies in CrCoNi Alloy Dec. 2021-Nov. 2022

Built Voronoi-initialized atomistic models, ran LAMMPS tensile simulations, and analyzed structural/strain features.

@ **IWM Institute, RWTH Aachen** | Supervisor: **Prof. Christoph Broeckmann** (c.broeckmann@iwm.rwth-aachen.de)

Finite-Element Modeling of Current-Assisted Sintering Nov. 2021-May 2022

Phase-Field Modeling of Sintering Sep. 2020-Feb. 2021

PUBLICATION

Tao Y, Cheng W and Wang W (2023) Atomistic tensile deformation mechanisms in a CrCoNi medium-entropy alloy with gradient nano-grained structure at cryogenic temperature. *Front. Mater.* 9:1118952. doi: 10.3389/fmats.2022.1118952

INTERNSHIP

Mercedes-Benz Group | Research & Development Intern | Untertürkheim, Stuttgart, Germany Jul. 2022-Jan. 2023

SKILLS

Software: Siemens NX, MATLAB, ABAQUS, Ansys HFSS, LAMMPS, MOOSE, OVITO, Origin, Blender, Keil, CubeMX

Programming language: C++, Python, MATLAB, Fortran, Java